8.3.4

EE24BTECH11013 - MANIKANTA D

Question:
Sketch the graph of $y = |x+3|$ and evaluate $\int_{-6}^{0} |x+3|\, dx$.
Solution:
Integral to calculate,

$$J = \int_{-6}^{0} |x+3|\, dx \tag{0.1}$$

Using the trapezoidal rule,

$$J = \int_{a}^{b} f(x)dx \approx h\left(\frac{1}{2}f(x_0) + f(x_1) + f(x_2) + \cdots + f(x_{n-1}) + \frac{1}{2}f(x_n)\right) \tag{0.2}$$

Recursive formula for the numerical solution:

$$J = j_n, \text{ where } j_{i+1} = j_i + h\frac{f(x_{i+1}) + f(x_i)}{2}, \tag{0.3}$$

$$x_{i+1} = x_i + h. \tag{0.4}$$

$$j_{i+1} = j_i + h\frac{|x_{i+1}+3| + |x_i+3|}{2}, \tag{0.5}$$

$$x_{i+1} = x_i + h. \tag{0.6}$$

where $h = \frac{b-a}{n}$, and $x_i = a + ih$.
Let $a = -6$, $b = 0$, and $f(x) = |x+3|$. Choose $n = 4$, so $h = \frac{0-(-6)}{4} = \frac{6}{4} = 1.5$.
The points are:

$$x_0 = -6, x_1 = -4.5, x_2 = -3, x_3 = -1.5, x_4 = 0 \tag{0.7}$$

The corresponding function values are:

$$f(x_0) = |-6+3| = 3, f(x_1) = |-4.5+3| = 1.5, \tag{0.8}$$

$$f(x_2) = |-3+3| = 0, f(x_3) = |-1.5+3| = 1.5, f(x_4) = |0+3| = 3 \tag{0.9}$$

Substitute into the trapezoidal rule formula:

$$J \approx 1.5\left(\frac{1}{2}f(x_0) + f(x_1) + f(x_2) + f(x_3) + \frac{1}{2}f(x_4)\right) \tag{0.10}$$

$$J \approx 1.5\left(\frac{1}{2}(3) + 1.5 + 0 + 1.5 + \frac{1}{2}(3)\right) \tag{0.11}$$

$$J \approx 1.5\,(1.5 + 1.5 + 0 + 1.5 + 1.5) \tag{0.12}$$

$$J \approx 1.5 \times 6 = 9 \tag{0.13}$$

The approximate value of the integral using the trapezoidal rule is $J \approx 9$.

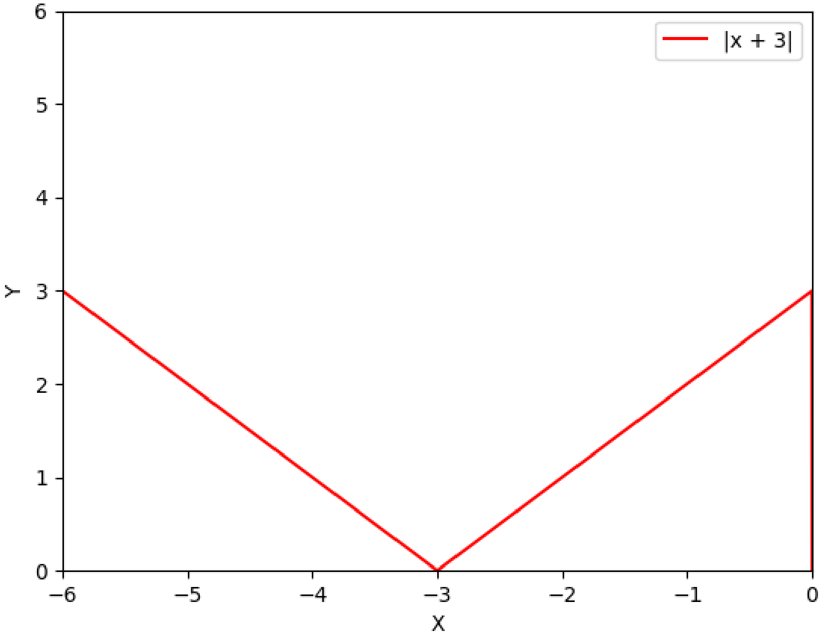

Fig. 0.1: Plot of the differential equation